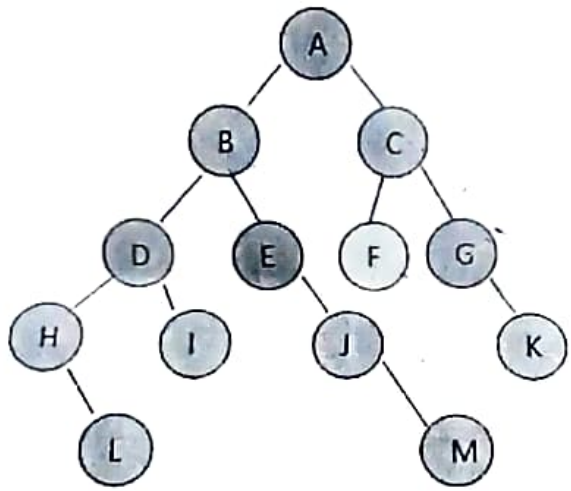
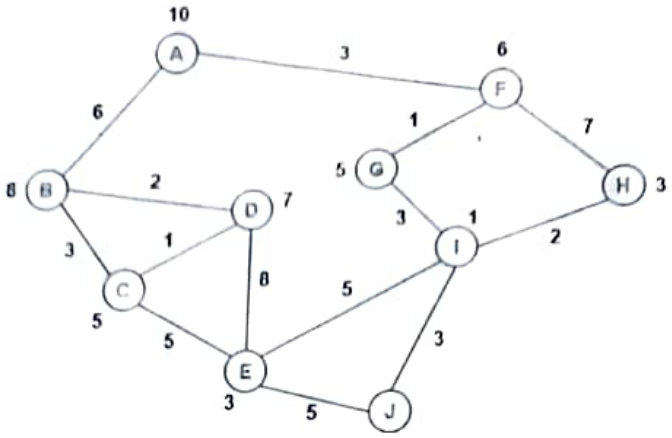
MID-TERM EXAMINATION
B.Tech. CSE-AI / AI&ML Semester: III
(October, 2023) OFF LINE mode

Subject Code: BAI-201	Subject: Artificial Intelligence
Time : 1 ½Hours	Maximum Marks : 30
Note:Q. 1 is compulsory.	

Q1		(2.5*4)	
	(a) Define intelligent agents, simplex reflex agents and goal-based agents.		
	(b) Explain the term artificial intelligence, and contribution of AI in various fields.		
	(c) Explain relevance of heuristic search in AI, and any one heuristic search technique.		
	(d) Define Structure of an Agent, with associated PEAS descriptions and examples.		
Q2	**(Attempt any Two Parts) UNIT-1**	(5,5)	
	a) Explain Breath First Search (BFS) and Depth First Search (DFS, and find the path for the following figure using BFS & DFS and highlight the properties of search techniques. A- Start node, F- Goal node. 		
	(b) Explain the reasons for preferring BFS technique over DFS, the limitations of DFS strategy, and compare uninformed search strategies with informed search strategies.		

(c) Explain the A* algorithm in AI with advantages and disadvantages. Consider the following graph-



The numbers written on edges represent the distance between the nodes.The numbers written on nodes represent the heuristic value.Find the most cost-effective path to reach from start state A to final state J using A* Algorithm.

Q3	(Attempt any Two Parts) UNIT-2	(5,5)
	(a) Explain bag of words (BoW) model. Why we use TF-IDF over BoW model?	
	(b) Define Knowledge representation in AI, the different techniques of Knowledge Representation, and the advantages of each technique.	
	(c) State first-order predicate logic? Explain syntax and semantics in Logical Representation, and the limitations of propositional logic.	